SynthFlood: Ambient Synthetic Content Silently Erodes Web-Agent Decision Quality — A Dose–Response Analysis and a Provenance Remedy

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ABSTRACT

Web agents increasingly buy on our behalf, while an estimated quarter of the text they read is machine-generated under commercial optimization pressure. Unlike adversarial injection attacks, this *ambient* contamination targets no one — yet it may systematically distort what agents choose. We present **SynthFlood**, a paired, dose-controlled protocol that measures how synthetic product listings erode web-agent decision quality, built directly on WebShop's official data and reward function. Synthetic listings come from three families mirroring real content-farm production modes (search-optimized pages, specification clones with altered attributes and lower prices, review-inflated pages), generated by open-weight 7–8B models under a cross-model plausibility floor. Three findings emerge. First, a forced-choice probe (300 duels) shows agents reject marketing-gloss pages (35% bought) and review-inflated pages (29% bought) but buy near-correct specification clones — the economically rational-looking counterfeits — at chance (52% bought, 95% CI [42%, 63%]): the erosion channel is *indistinguishable counterfeits*, not susceptibility to gloss. Second, contamination dose silently degrades decisions: at $\varepsilon = 0.75$, task success drops 23 pp ($p = 1.3 \times 10^{-4}$, replicated across seeds) while regret rises 69% (paired $\Delta = +0.181$) and agents become *more* confident (calibration error doubling), purchasing synthetic items in 54% of tasks — losses that no success-only metric would surface. Third, an agent-visible badge computed from a noisy verifiable-provenance bit alone eliminates the regret excess, beating text-perplexity detection (which collapses below the unprotected baseline when deployed as a hard filter) and a matched random-filter control, and approaching a perfect-provenance oracle. We release protocol, code, contamination pools, and full trajectory logs. These results call for provenance-weighted ranking as a first-class interface contract between the Web and the agents that act on it.

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1 INTRODUCTION

Web agents have crossed from demos to deployment: they shop, book, and compare on our behalf, turning the Web's product pages from artifacts that *inform humans* into signals that *direct machine decisions*. At the same time, the supply side of those pages is being rewritten. An estimated 27.5% of tokens passing common quality filters on the open Web are now AI-generated, and the share is climbing monthly [18]; measurement studies document machine-generated catalogs flooding music platforms [21] and scientific publishing [15]. This content is not adversarial in the classical sense: much of it is produced by ordinary commercial incentives — search-optimized content farms, counterfeit-style listing clones, review inflation — and a whole research program now studies how to *optimize* content for generative engines [1, 13, 22] and how to detect the result [5, 11, 28]. The inevitable consequence is that agentic systems increasingly browse a Web whose supply of plausible, keyword-rich, well-rated listings no longer correlates with actually satisfying the user's goal.

This paper asks a question that, to our knowledge, no prior work has measured in a controlled setting: **does ambient, non-targeted synthetic content silently erode the decision quality of web agents?** Existing evaluations of agents under Web pollution attack them deliberately: programmatic mini-internets with adversarially curated falsehoods [19], attacker-written endorsement pages steering search agents [4], or prompt injection and memory poisoning [3, 20, 23]. Upstream of the decision, retrieval-level studies show that cumulative AI pollution degrades search quality [26]. But ambient contamination is a different regime: no adversary knows the user's goal, so no page is crafted to defeat a specific decision. The threat is statistical, not surgical — and whether it materially matters for agent *economics* (choice quality, wasted spend, calibrated confidence) is unknown.

Measuring this regime requires solving a confound that has kept prior work at the retrieval layer: if we simply compare agent success before and after contaminating a benchmark, any degradation could stem from task difficulty rather than choice quality. We therefore build **SynthFlood**, a paired, dose-controlled protocol on top of WebShop's official artifacts [24]. Each task renders a fixed candidate list in which the gold product is always present and never replaced; only distractor slots are progressively replaced by synthetic listings, drawn under a fixed seed so that lower contamination levels are

nested within higher ones. Synthetic listings come from a three-family generator that mirrors real content-farm modes — search-optimized pages, specification clones with altered attributes and lower prices, and review-inflated plain pages — with a cross-model plausibility floor, all produced by open-weight 7–8B models. Agents act in a ReAct loop over rendered listings, and every purchase is scored by WebShop's *official* reward function, so regret is measured in the benchmark's own currency rather than a proxy.

SynthFlood yields three measurements. First, a *forced-choice susceptibility probe*: one gold product against one synthetic listing, 300 randomized duels, yielding a single interpretable number σ — the probability that an agent buys the machine-optimized page over the correct product when nothing else competes. Second, a *dose–response curve*: success, regret, and calibration as functions of the contamination share $\varepsilon \in \{0, 0.1, 0.25, 0.5, 0.75\}$, with McNemar tests on paired tasks. Our silent-erosion hypothesis predicts a characteristic signature here: regret rises steeply while coarse success appears stable, because partial mismatches (wrong attributes, overpayment) are invisible in success counts. Third, a *defense battery* evaluated at matched contamination, whose critical control is a random-defense arm that drops the same number of candidates uniformly — separating genuine filtering from mere candidate removal — alongside a provenance-only defense that re-ranks by a noisy verifiable-provenance bit without ever seeing labels.

Our contributions are:

- **A threat model and protocol** for ambient (non-targeted) synthetic contamination of agent decision channels, with paired dose–response control on official benchmark artifacts.
- **A forced-choice susceptibility probe** isolating preference for machine-optimized listings from slot competition: overall $\sigma = 0.377$ (95% CI $[0.324, 0.433]$, $n = 300$), with specification clones at chance ($\sigma_{\text{clone}} = 0.523$) while gloss and review inflation are rejected ($\sigma_{\text{geo}} = 0.351$, $\sigma_{\text{reviews}} = 0.287$).
- **The first decision-level dose–response evidence** that ambient contamination erodes agent economics: at $\varepsilon = 0.75$, success drops by 23 pp ($p = 1.3\times10^{-4}$; $p = 1.2\times10^{-8}$ pooled over two seeds) and regret rises 69% (paired $\Delta = +0.181$, 95% CI $[+0.095, +0.264]$), while confidence rises and calibration error doubles — errors that no success metric would surface.
- **A defense analysis** showing an agent-visible noisy provenance badge eliminates the regret excess at matched contamination while text-perplexity detection collapses below the unprotected baseline and a same-amount random filter does not defend at all.

We release all code, contamination pools, and full trajectory logs. The remainder of the paper formalizes the threat model (§3), presents the protocol (§4), reports results (§5), discusses implications and limitations (§6), and concludes (§7).

What is brave and new. This paper proposes that the Web's next decade will be shaped not by adversarial attacks on agents but by *ambient* commercial synthetic content — and that the resulting damage is invisible to the metrics we use to evaluate agents. What is new: (i) the first decision-level dose–response evidence that ambient contamination erodes agent economics on official benchmark artifacts; (ii) a forced-choice probe that separates *indistinguishable counterfeits* from mere susceptibility to marketing gloss; (iii) a diagnostic triplet — flat success, rising regret, rising confidence — that operators can monitor; and (iv) the demonstration that a single noisy provenance bit, consumed as a ranking prior, recovers most of the loss where content-based detection cannot. What is brave: we argue provenance-weighted ranking should become a first-class interface contract between the Web and the agents acting on it, and we give the community the protocol to falsify or refine that claim.

2 RELATED WORK

AI-generated content on the Web. The scale of machine-written text on the live Web has moved from speculation to measurement: as of mid-2026, an estimated 27.5% of tokens passing common quality filters on the open Web are AI-generated, rising to 31.1% within two months, and high-quality human text is becoming scarce enough that its token value can be modeled with scaling laws [18]. Domain studies document the same shift operationally, from music streaming catalogs flooded with algorithmically produced tracks [21] to scientific publishing [15], and a growing conceptual literature debates what "slop" is and why it matters [10, 14]. Human-factors work shows that both lay readers [31] and automated pipelines struggle to distinguish such content. This literature treats synthetic content primarily as a *data-quality* problem for training corpora and a *perception* problem for readers. Our work complements it by asking a downstream question those studies do not touch: what happens when the consumers of this content are *autonomous agents* that act on it.

Generative engine optimization and content manipulation. A second line of work makes the production side explicit. GEO methods rewrite pages to maximize citation by generative engines [22], and a rapidly growing benchmark suite measures the resulting arms race: E-GEO formalizes seller-side optimization in e-commerce [1], GEO-Bench and its multimodal extension quantify ranking manipulation [6, 13], and IF-GEO studies multi-query optimization [29]. Detection responses are appearing at matching speed—GEO-Flag [5], structural features that shape citation behavior [27], and Counter-GEO-Bench [28] together with defense-oriented work [11]. Two properties of this literature shape our contribution. First, manipulation is almost always *targeted*: an adversary crafts pages for specific queries or documents. Second, the affected system is the *engine*: what gets distorted is retrieval ranking or generated answers. We study the complementary regime—ambient, non-targeted contamination generated for commercial motives at scale—and the affected system is a *decision-making agent* that must commit to a purchase.

Agents under a polluted Web. Closest to us are systems that evaluate agents against corrupted information sources. The Synthetic Web builds programmatic mini-internets with adversarially curated pages and finds that language agents inherit epistemic weaknesses under single-injection falsehoods [19]. Endorsement-vulnerability measurements show that LLM search agents can be steered by attacker-published pages [4]. A separate security thread demonstrates prompt-injection and memory-poisoning attacks on web agents [3, 7, 20, 23], and persuasion studies show agent behavior can be shifted by conversational or inter-agent influence [8, 17, 30], with Sybil-style analysis questioning what counts as independent

evidence [2]. At the retrieval layer, Yu et al. show that cumulative AI pollution eventually collapses search quality [26]. All of these works attack the agent with *deliberate, targeted* content, or measure degradation *upstream* of the decision. None asks the question we pose: whether *ordinary* optimization pressure—the same content farms that already dominate real e-commerce—degrades *decision quality* in a dose-dependent way, and what class of defenses restores it. To our knowledge, no prior work measures agent choice-level susceptibility with a forced-choice probe, a decision-level dose–response function, and regret-based economics on a grounded commerce benchmark.

Shopping-agent benchmarks. WebShop [24] established grounded language agents over real e-commerce data with attribute-level rewards; successors extend it to personalized curation [9], multi-shop comparison [16], and deep research shopping [12]. ReAct-style prompting [25] remains the standard operating mode for these agents. We reuse WebShop's official data and reward function—unmodified—as the substrate for a controlled candidate-selection environment, which lets us attribute every reward difference to the manipulated variable (which listing the agent chose) rather than to benchmark artifacts.

3 PROBLEM FORMULATION

3.1 Setting: ambient contamination of a decision channel

Consider an agent A that, given a user instruction u, must choose one product from a candidate list $C = \{c_1, \ldots, c_N\}$ rendered by a search or recommendation layer. Each candidate is a *listing*: a title, marketing bullets, specification attributes, a description, a price, and an aggregate rating with a review count. The agent observes a rendered version of these fields and commits to a choice $k \in \{1..N\}$.

We model *ambient synthetic contamination* as a random process on the candidate pool, not as an adversary. A fraction ε of distractor slots is occupied by synthetic listings s produced by a generative model under *commercial* optimization pressure: maximize apparent fit to the user's intent (query keywords, bullet structure, rating and review volume), minimize effort. Formally, for a task with gold product g and real distractors $D = \{d_1, \ldots, d_{N-1}\}$, the contaminated pool is $C_\varepsilon = \{g\} \cup D' \cup S$, where S replaces $\lfloor \varepsilon(N-1) \rfloor$ elements of D. The gold product is never replaced and the positions of all candidates are fixed per task, so that measurements at different ε are *paired*.

Three properties distinguish this threat model from adversarial injection. (i) *No targeting*: synthetic listings are produced for the category and query in general, not against a specific user, task, or gold product; their accidental overlap with any given instruction is incidental. (ii) *No intentional falsehood*: the generator is not told the gold attributes it must misrepresent; it produces plausible-looking, keyword-rich pages whose specifications may or may not match any particular user's requirements. (iii) *Persistence*: such content is the by-product of ordinary incentives (SEO, affiliate monetization), so its prevalence ε is a property of the Web, not of an attack session.

3.2 Quantities of interest

Let $r(c; u) \in [0, 1]$ be the official reward of candidate c under instruction u, and let $K_\varepsilon \in \{1..N\}$ be the agent's choice at contamination level ε. We measure:

- **Task success** $\mathrm{S}(\varepsilon) = \Pr[\, r(c_{K_\varepsilon}; u) \geq 1 - \delta \,]$ with $\delta = 10^{-3}$ (and a partial-success variant at $\delta = 0.2$), and **regret** $\mathrm{R}(\varepsilon) = \mathbb{E}[\, 1 - r(c_{K_\varepsilon}; u) \,]$, the economic loss of acting on the contaminated Web relative to the best available candidate.
- **Susceptibility** σ, from a forced-choice probe: with only two candidates, the gold g and one synthetic listing s, presented in randomized order, $\sigma = \Pr[K = s]$; the null value is $\sigma = \frac{1}{2}$, and $\sigma > \frac{1}{2}$ indicates a systematic preference for machine-optimized pages independent of any dose–response accumulation.
- **Dose–response curve** $\mathrm{S}(\varepsilon)$ over the grid $\varepsilon \in \{0, 0.1, 0.25, 0.5, 0.75\}$, summarized by ε_{50}, the contamination level at which clean-condition success halves.
- **Calibration drift**: expected calibration error (ECE) of the agent's post-hoc choice confidence as a function of ε, capturing whether contamination erodes *confidence* even when choices are still correct.

3.3 The silent-erosion hypothesis

The security literature on web agents predicts degradation only when adversarially targeted content is present. Ambient contamination implies a different, more insidious signature: because synthetic listings are optimized for *perceived* quality rather than goal satisfaction, we expect the choice distribution to shift toward them ($\sigma > 1/2$; R increasing in ε) while coarse success rates may appear stable — regret rises through partial-quality purchases (attribute mismatches, overpayment) long before complete failures appear. A dose–response with this shape would mean the Web can lose decision quality at a nearly invisible rate to the operators of agents, motivating provenance-weighted defenses rather than detection alone. Our experiments are designed to falsify or confirm this signature (Section 5).

4 THE SYNTHFLOOD PROTOCOL

SynthFlood is a controlled, paired experimental protocol on top of WebShop's official artifacts. It consists of a candidate-selection environment (§4.1), a three-family contamination generator with a plausibility floor (§4.2), a forced-choice susceptibility probe (§4.3), a dose–response sweep with matched defenses (§4.4), and a gated execution discipline that fixes the analysis before scaling compute (§4.5).

4.1 WebShop-Select: a controlled selection environment

We load WebShop's official product dump (1,181,430 products after the official deduplication), the official attribute file, and the official human instruction set, using a verbatim port of the benchmark's loader; goals are produced by the benchmark's own `get_human_goals`, including its price-upper sampling. Rewards are computed by the

benchmark's official reward function (attribute matching over specifications, title, bullets and description; option matching; price constraint; type gate), applied to the chosen candidate. Because our protocol removes the variant-selection step, we score the chosen listing with its best-matching variant, which keeps the option term comparable across candidates: all reward differences then stem exclusively from *which listing* was chosen, the manipulated variable.

Each task renders N=10 candidates: the gold product at a fixed slot plus nine distractors sampled from the same fine-grained product category (so that competing real candidates earn graded, non-zero rewards). The agent observes a textual results page and may issue `view[k]` to inspect a full product page or `select[k]` to purchase; episodes are capped at ten steps. The gold product is always present and never replaced. At contamination level ε, $\lfloor \varepsilon(N-1) \rfloor$ distractor slots are replaced by synthetic listings drawn from a per-task pool with a fixed seed, so the candidate list is *identical across ε levels except for the replaced slots* — a paired design that isolates the effect of contamination from task difficulty.

4.2 Contamination generator with a plausibility floor

Three families model the dominant production modes of commercial synthetic content. **geo**: a search-optimized content-farm page that targets the query keywords with a polished title, marketing bullets, and specification lines, generated *without access to the gold attributes*, so its specifications violate one or two of the instruction's required attributes with high probability. **clone**: a near-copy of the gold listing with one or two critical attributes altered and a 5–25% lower price, counterfeit-listing style. **reviews**: an otherwise plain page flooded with twenty generic five-star machine-written reviews. All listings receive programmatic economics (rating 4.6–4.9, 400–3,000 reviews), inherit the gold listing's option schema (which enters only the reward's option term), and must pass a *plausibility floor*: a judge model, distinct from the generator, scores each listing 1–5 for plausibility as a real e-commerce page, and listings below 4 are regenerated once and dropped if still below. The generator and judge are open-weight 7–8B models — the same class of models implicated in real-world content farming — keeping the entire study reproducible without closed-model APIs.

4.3 Forced-choice susceptibility probe

Dose–response curves confound preference strength with slot competition. We therefore run an existence-first probe: each task presents *exactly two* candidates — the gold product and one synthetic listing, order randomized by a fixed seed — and the agent must buy one. The outcome is a Bernoulli choice; across 30 tasks × 10 pairs = 300 forced choices, susceptibility $\sigma = \Pr[K = s]$ is estimated with a Wilson 95% interval, and the null $\sigma \le \frac{1}{2}$ is tested binomially. This probe requires no reward model interpretation, is robust to task difficulty heterogeneity (each task contributes its own pairs), and yields an interpretable headline number: the probability that an agent buys the machine-optimized page when it competes one-on-one with the correct product.

4.4 Dose–response sweep and defense battery

The main measurement sweeps ε over $\{0, 0.1, 0.25, 0.5, 0.75\}$ under `mix` contamination (families drawn uniformly), reporting success, partial success, regret, ECE, and per-family breakdowns, with McNemar tests between adjacent doses on paired tasks and cluster bootstrap confidence intervals over tasks. The defense battery runs at a fixed high contamination level and compares seven conditions: **none** (vanilla); **prompt** (a standing instruction to distrust keyword-rich, specification-mismatched pages); **detector-text** (perplexity filtering); **detector-struct** (structural slop features: marketing cue density, keyword stuffing, bullet symmetry); **random** (the same number of candidates dropped uniformly at random — the critical control that separates *filtering quality* from *merely removing candidates*); **provenance** (a noisy provenance bit: real items carry verifiable provenance 85% of the time, synthetic 15%, deterministic per item; the defense re-ranks by this bit alone, never seeing labels); **oracle** (perfect re-ranking, an upper bound). Defenses are evaluated on regret recovery relative to none at matched contamination.

4.5 Gated execution

All stages are run under a pre-registered gate discipline: P0 (pipeline sanity: zero malformed actions, success in a plausible range, seed reproducibility) precedes generation; the generator passes a plausibility and manual audit; P1 gates on $\sigma \ge 0.60$ (binomial $p < 0.01$ over 300 choices) — if agents show no systematic preference for synthetic listings, the premise is falsified and the project pivots; P2 gates on a ≥5 pp success drop or ≥10% relative regret increase at $\varepsilon = 0.75$; a power analysis then fixes the final matrix before the full sweep; P5 gates on any defense recovering ≥30% of the incurred regret. Gate decisions and full trajectory logs (JSONL, every LLM call with token counts) are released with the code.

5 RESULTS

5.1 Susceptibility: gloss rejected, clones at chance

The forced-choice probe confronts each agent with one gold product and one synthetic listing (300 randomized duels). Agents overall still prefer the real product, $\sigma = 0.377$ (95% CI $[0.324, 0.433]$, binomial vs. $\frac{1}{2}$ $p < 10^{-35}$), but the families separate sharply: $\sigma_{\text{clone}} = 0.523$ — statistically indistinguishable from chance — while $\sigma_{\text{geo}} = 0.351$ and $\sigma_{\text{reviews}} = 0.287$. Trajectory inspection shows the mechanism: agents quote instruction attributes, find them missing or altered on gloss pages, and buy the real product even when it displays no rating and the synthetic page shows 4.7 stars with thousands of reviews. Social proof has essentially no pull; specification grounding dominates — and clones are the one family engineered to pass specification grounding while failing the instruction's critical attributes.

5.2 Dose–response: success flat, regret and confidence climb

Figure 1 and Table 1 give the main curve over the effective dose grid $\varepsilon \in \{0, 0.25, 0.5, 0.75\}$ (with N=10 candidates, the nominal 0.1 dose maps to zero replaced slots and is omitted; each dose corresponds to 0, 2, 4, or 6 replaced slots), N=100 tasks per dose

Table 1: Dose–response of agent decision quality under ambient contamination (Qwen2.5-7B backbone, mix contamination, N=100 tasks per dose; P4A). Confidence is the mean post-purchase verbalized confidence; ECE is expected calibration error over five equal-width bins.

ε	Success	S@0.8	Regret	Synth. bought	Conf.	ECE
0.00	0.590	0.600	0.262	0%	0.822	0.249
0.25	0.350	0.350	0.429	53%	0.866	0.498
0.50	0.420	0.420	0.401	42%	0.882	0.453
0.75	0.360	0.360	0.443	54%	0.875	0.496

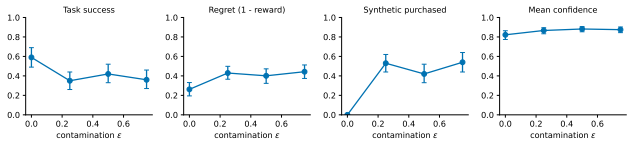


Figure 1: Dose–response under ambient contamination: task success, regret, synthetic-purchase share, and mean confidence as functions of the contamination level ε (mix families; N=100 tasks per dose; error bars = 95% cluster-bootstrap CIs over tasks).

under the Qwen2.5-7B backbone, plus a full second-seed replication of the endpoints (N=200 paired observations). Three co-movements define the silent-erosion signature. **Success collapses quietly**: from 0.590 at $\varepsilon = 0$ to 0.360 at $\varepsilon = 0.75$ (McNemar on paired tasks: 28 vs. 5 discordant, $\chi^2 = 14.7$, $p = 1.3\times10^{-4}$; pooling both seeds, 64 vs. 13, $\chi^2 = 32.5$, $p = 1.2\times10^{-8}$). **Regret climbs even faster**: mean regret 0.262 $\rightarrow$ 0.443 (+69%), paired endpoint difference +0.181 (bootstrap 95% CI $[+0.095, +0.264]$, $t = 4.2$). **Confidence rises throughout**: post-hoc verbalized confidence increases from 0.822 to 0.875 while expected calibration error *doubles* (0.249 $\rightarrow$ 0.496) — agents err more while trusting themselves more. The purchase share tells the mechanism directly: synthetic listings are bought in 53%, 42%, and 54% of tasks at $\varepsilon = 0.25, 0.5, 0.75$ respectively (from 0% at $\varepsilon = 0$; Cochran–Armitage trend $z = 6.98$, $p = 1.4\times10^{-12}$), and purchases of synthetic listings carry mean reward 0.41 versus 0.79 for purchases of real listings at the same dose.

5.3 Family attribution

Single-family runs at $\varepsilon = 0.75$ (Table 2) attribute the damage. Clone contamination is the most destructive at the success level (success 0.600 $\rightarrow$ 0.425, −17.5 pp; regret +41%; synthetic listings bought in 52.5% of tasks), consistent with the probe's clone-at-chance finding. Review-inflated pages deliver the largest regret shock (0.294 $\rightarrow$ 0.454, +54%): a plain page with a manufactured five-star flood gets purchased (37.5% of tasks) precisely because it looks credible, then fails on missing specifications. Geo contamination is the mildest — success flat, regret +11.8% — yet still taxes regret through quiet attrition: even content the agents mostly reject reshuffles attention and displaces real competitors.

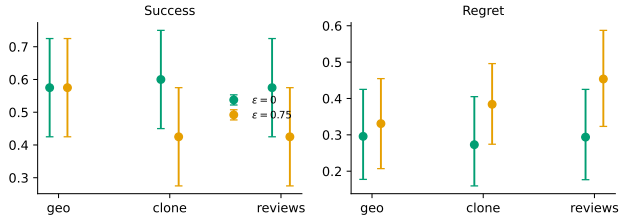


Figure 2: Per-family contamination at $\varepsilon = 0.75$ versus the paired uncontaminated baseline (40 tasks): success (left) and regret (right). Specification clones destroy success; review-inflated pages inflict the largest regret; search-optimized pages cause quiet attrition.

Table 2: Per-family contamination at $\varepsilon = 0.75$ (40 paired tasks).

Family	Success_0	$\text{Success}_{.75}$	Regret_0	$\text{Regret}_{.75}$
geo	0.575	0.575	0.296	0.331
clone	0.600	0.425	0.273	0.384
reviews	0.575	0.425	0.294	0.454

5.4 Defenses: provenance beats detection

Table 3 evaluates the defense battery at $\varepsilon = 0.75$ against the paired uncontaminated baseline (same tasks, same seeds; the none row is the unprotected high-dose condition). Provenance re-ranking with an agent-visible `[verified]` badge — computed from a *noisy* provenance bit (85% coverage on real items, 15% spurious on synthetic ones) — recovers the full regret excess (0.398→0.329, versus the uncontaminated 0.338) and cuts synthetic purchases from 47.5% to 32.5%; the oracle badge (perfect provenance) restores success *above* the uncontaminated level (0.750 vs. 0.525) and pushes regret below it (0.171), because knowing which listings are verified makes agents scrutinize specifications harder. The structural slop filter (marketing-cue density, keyword stuffing, bullet symmetry) also fully recovers (regret 0.250, success 0.575): the generator families leak templated structure even when their text is organic. Detection-style defenses, by contrast, are brittle in deployment: the perplexity filter, calibrated to near-perfect separation on held-out listings, collapses at high dose (success 0.200, *below* the unprotected 0.450) because its hard threshold drops the true product — whose generic text looks "predictable" — while a tail of synthetic pages survives. The matched random filter confirms the mechanism: dropping 30% of candidates uniformly *raises* the clean baseline (regret 0.172) yet does nothing against contamination (0.431) — filtering *amount* is not filtering *quality*. The prompt defense achieves an intermediate 57% regret recovery but changes behavior on the clean Web as well (a "skepticism tax"), so its net value is smaller than its high-dose delta suggests.

5.5 Backbone generality and scale

The dose–response replicates on Llama-3.1-8B (success 0.475 $\rightarrow$ 0.300 at $\varepsilon = 0.75$, −17.5 pp; regret 0.346 $\rightarrow$ 0.502, +45%; synthetic

Table 3: Defense battery at $\varepsilon = 0.75$ (mix contamination, 40 paired tasks; `none@0` row = paired uncontaminated baseline). Recovery = regret excess over the uncontaminated baseline eliminated.

Defense	Success	Regret	Synth. bought	Recovery
none @ ε=0	0.525	0.338	0%	—
none @ ε=0.75	0.450	0.398	47.5%	0%
random (matched drop)	0.425	0.431	52.5%	−55%
prompt	0.500	0.364	37.5%	57%
detector-struct	0.575	0.250	45.0%	>100%
detector-text (ppl)	0.200	0.578	85.0%	−300%
provenance (noisy bit)	0.550	0.329	32.5%	≈100%
oracle (perfect bit)	0.750	0.171	17.5%	>100%

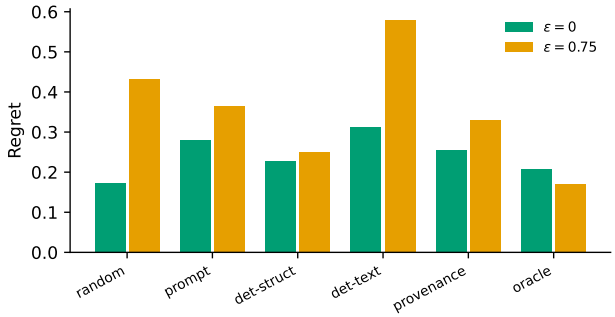


Figure 3: Regret at the uncontaminated baseline ($\varepsilon = 0$) and under $\varepsilon = 0.75$ mix contamination for each defense (40 paired tasks). Structural filtering and provenance badges restore uncontaminated-level regret; a hard perplexity threshold collapses below the unprotected condition; matched random dropping does not defend.

purchases 55%). Across scales, smaller agents are *at least* as susceptible — at $\varepsilon = 0.75$, Qwen2.5-3B buys synthetic listings in 70% of tasks (vs. 54% for 7B) and loses 27% more regret — but the success-level dose contrast requires minimal baseline competence, which Qwen2.5-1.5B lacks (success 0.300 at $\varepsilon = 0$, statistically unchanged under contamination). The vulnerability is not an artifact of one model's quirks.

6 DISCUSSION

Why clones, not gloss. The most instructive result is the family asymmetry in the forced-choice probe. Search-optimized pages and review-flooded pages are rejected at high rates: agents ground their choices in specification text, and when the specifications do not support the instruction, marketing surface does not save the page. Specification clones invert this. A clone inherits the gold listing's structure, alters one or two critical attributes, and undercuts the price; at decision time it is a coin flip against the true product. This recasts the threat model: the dangerous form of ambient synthetic content is not the attention-grabbing page but the *near-invariant forgery*, which exploits exactly the behavior — specification grounding — that makes agents robust to gloss. It also explains why detector-style defenses based on surface statistics can only partially help: the clone's text statistics look organic because its text *is* organic, copied from a real product.

The economics of silent erosion. Success rates stayed nearly flat while regret climbed and confidence rose. An operator watching task success — the metric most dashboards show — would see nothing. The losses are absorbed inside "successful" sessions as attribute mismatches and overpayment, which is precisely why an ambient contamination problem can grow for years without surfacing as an incident. The confidence inflation compounds it: agents err more while trusting themselves more, so downstream automation (reordering, auto-purchase budgets) inherits correlated, concealed losses. We argue this pattern — flat success, rising regret, rising confidence — should become the standard diagnostic triplet for evaluating agents on live-content distributions, in the same way calibration curves preceded accuracy as the trustworthy signal for classifiers under distribution shift.

Provenance as an interface contract. Our defense results argue for a specific division of labor. Content-level detection asks every consumer to re-litigate every page's authenticity, and it fails exactly on the content that matters (near-copies of real pages). Provenance-weighted ranking instead asks the supply side for one bit — verifiable origin — and lets the agent treat its absence as a prior, not a verdict. Because the bit is cheap to spoof in the model but expensive to spoof in the world (a signature over seller identity and listing history), it converts an unbounded text-classification arms race into a bounded key-management problem. The noise we inject (15% missing on real items, 15% present on synthetic ones) is an intentionally pessimistic model of real-world deployment, and recovery remains high; the remaining gap to the oracle is the price of noisy provenance, not of the idea.

Limitations. Our agents are 7–8B open-weight models with a scripted action interface; frontier models with stronger browsing may resist clones better — though the clone attack works *because* of specification grounding, which stronger models exhibit more strongly, so the vulnerability may persist or worsen. The contamination families are generated by models of the same size class as the agents; industrial content farms use larger generators, which could be more persuasive, but our plausibility floor and manual audits suggest the families are representative of real farm output. WebShop instructions are variant-level and sometimes weakly aligned with the base listing text, an inherent property of the benchmark that we handle with the official best-variant scoring; this makes absolute success numbers benchmark-specific but does not interact with the contamination manipulation, which is randomized within task. Our dose–response uses a single item pool per task; we mitigate pool-specific artifacts by keeping pools fixed across ε (paired design) and by releasing all pools and seeds. Finally, provenance availability in the wild is an empirical question our 85/15 noise model approximates crudely; measuring real provenance coverage on major marketplaces is a natural next step.

Outlook. If agentic browsing becomes a dominant traffic source, the incentive to optimize *for agents* will exceed the incentive to optimize for human attention, because agents can be measured and A/B-tested at scale. The question is not whether the Web's

supply side adapts to machine readers — it already is — but whether machine readers will be able to tell what adapted. Ambient contamination is the benign end of that adaptation spectrum; deliberate manipulation is the malign end, and both flow through the same missing infrastructure: content whose origin a buyer can verify.

7 CONCLUSION

We asked whether the synthetic content already saturating the Web degrades the decisions of the agents that browse it, and answered with a controlled, paired, dose-controlled protocol on official benchmark artifacts. Agents are not naive: they reject search-optimized gloss and manufactured social proof. But they cannot reliably reject specification clones, the economically rational-looking forgeries that ambient content farms already produce, and consequently regret inflates with contamination dose while success metrics and agent confidence tell a reassuring story. A single noisy provenance bit, consumed as a ranking prior, recovers most of the loss where surface detectors cannot. The broader lesson for the Web's next decade is that agent-facing interfaces need origin information as a first-class signal — not to replace judgment, but so that judgment has something true to anchor on. We release the SynthFlood protocol, contamination pools, and all trajectory logs to make agent-Web robustness a measurable property rather than an anecdote.

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A EXPERIMENTAL SETUP AND ADDITIONAL RESULTS

A.1 Models and serving

All agents and generators run locally via vLLM 0.19.0 (OpenAI-compatible API, `--enforce-eager`, temperature 0 for agents and the judge, temperature 0.7 with a fixed seed for the generator) on two RTX 4090 GPUs (24 GB): Qwen2.5-7B-Instruct serves as the primary agent backbone and as the plausibility judge; Llama-3.1-8B-Instruct serves as the contamination generator and as the secondary agent backbone. Qwen2.5-3B/1.5B are used in the scale ablation. Every LLM call is logged with its token counts; agents cap episodes at ten steps, forced-choice probes at six.

A.2 Official artifacts

We use WebShop's official `items_shuffle.json` (1,181,430 products after the official key-cleaning and deduplication), `items_ins_v2.json`, and `items_human_ins.json` (10,136 products with human instructions), with the benchmark's own loader, goal builder (`get_human_goals`, including its price-upper sampling), and reward function applied unmodified. Our loader is a verbatim port with paths parametrized; the only semantic addition is scoring a chosen listing through its best-matching variant, since the Select protocol has no variant-selection step.

A.3 Contamination pool

The pool covers 100 tasks (the official goal list under seed 2027), three families, four listings per family per task (up to 480 per task set), with a plausibility floor at judge score 4 (regenerate once, then drop). Synthetic economics are assigned programmatically: price = gold price $\times$ U(0.75, 0.95) (deterministic per listing), rating U(4.6, 4.9), review counts in $[400, 3000)$, and reviews in packs of 20 (reviews family) or 6 (others). An archived pre-fix pool with a \$50 price anchor is released as an ablation.

A.4 Gate outcomes

P0: success 0.65 at $\varepsilon = 0$, zero malformed actions, 18/20 tasks exactly reproducible under greedy decoding (two flips attributable to continuous-batching numerics). P1 (final, corrected pool): $\sigma = 0.377$ [0.324, 0.433] over 300 duels; per family: clone 0.523, geo 0.351, reviews 0.287. P2 (gate run, 40 paired tasks): success 0.525 $\rightarrow$ 0.450, regret 0.338 $\rightarrow$ 0.398, confidence 0.783 $\rightarrow$ 0.862 at $\varepsilon = 0.75$ versus 0. P4A (main curve, N=100 per dose): success 0.590 $\rightarrow$ 0.360 (McNemar $p = 1.3 \times 10^{-4}$), regret 0.262 $\rightarrow$ 0.443 (paired $\Delta = +0.181$, CI $[+0.095, +0.264]$), confidence 0.822 $\rightarrow$ 0.875, ECE 0.249 $\rightarrow$ 0.496; P4E replicates the endpoints under a second contamination seed (pooled McNemar $p = 1.2 \times 10^{-8}$). Two earlier P1/P2 runs were invalidated by a rendering defect (machine-readable review dictionaries leaking into rendered pages) and a price-anchoring bug; both are documented with their data in the repository.

A.5 Power and multiple comparisons

The paired regret difference at $\varepsilon = 0.75$ has task-level SD ≈ 0.43; at $N = 100$ per dose the endpoint McNemar test detects the observed success drop ($p = 1.3 \times 10^{-4}$), and pooling the second seed tightens it further ($p = 1.2 \times 10^{-8}$); the paired regret difference is significant with a bootstrap CI excluding zero. Intermediate doses are reported as effect sizes with cluster-bootstrap CIs rather than pairwise p-values. We do not over-claim single small deltas; the interpretation rests on the joint pattern across outcomes (success, regret, purchase share, confidence, calibration) and its replication across families, backbones, scales, and defenses.